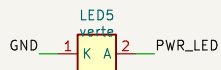
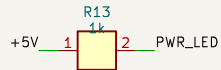
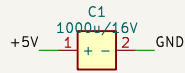
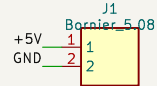
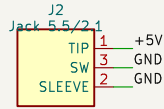
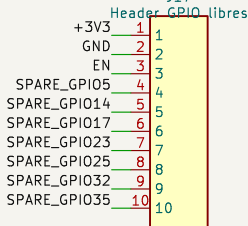


GPIO2/GPIO15 = broches de strapping : pull-down base 10k = état sûr au boot.

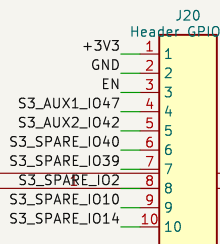
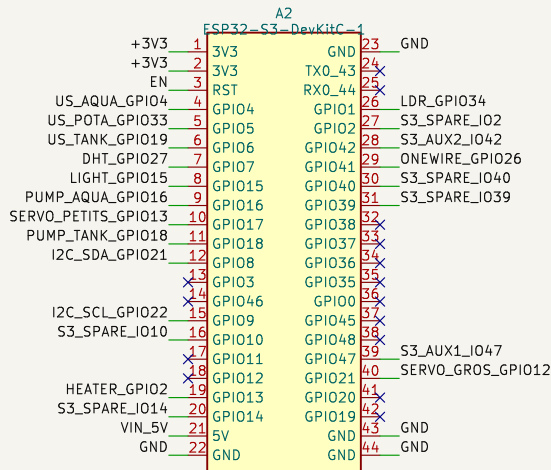
ALIMENTATION 5V (3A recommandé) – jack OU bornier.
D5 protège le rail si l'USB du DevKit est branché en même temps.



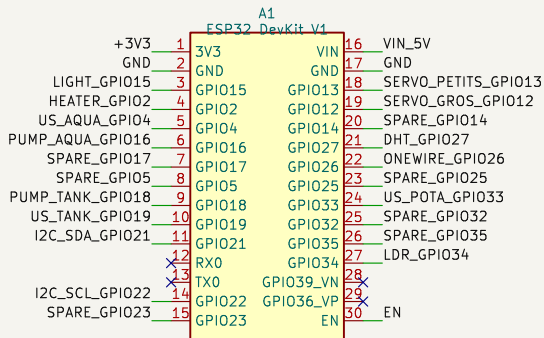
GPIO libres pour évolutions.



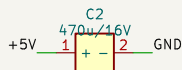
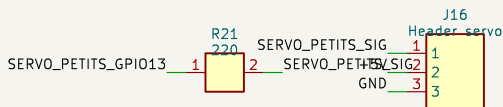
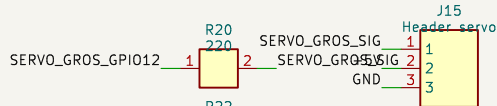
SITE A2 (option) : ESP32-S3-DevKit-1, cartographie PINMAP_S3_CARRIER (pins.h). UN SEUL module peuplé : A1 (WROOM) OU A2 (S3).
Mêmes nets fonctionnels que A1 ; J20 = breakout GPIO S3 (AUX1=IO47, AUX2=IO42).



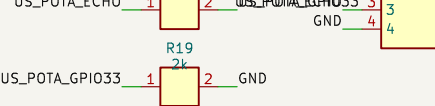
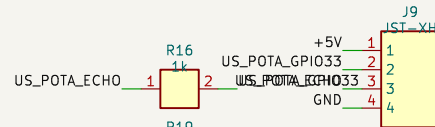
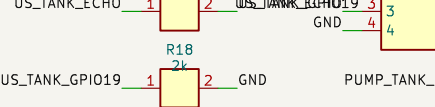
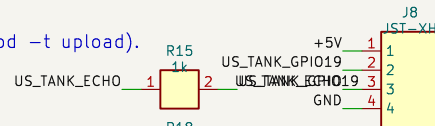
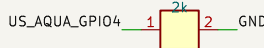
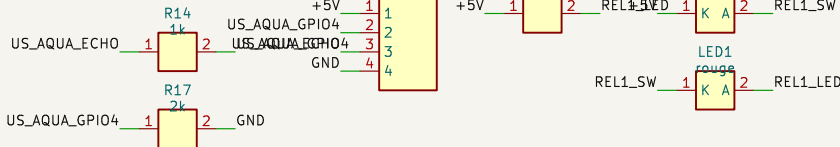
ESP32 DevKit V1 (30 broches, socketé).
Flash/monitor via l'USB du module (pio run -e wroom-prod -t upload).



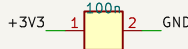
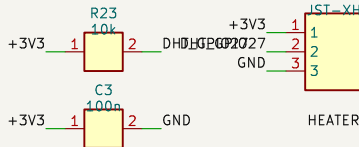
SERVOS NOURRISSEURS 5V
GPIO12 (strapping MTDI) : pull-down R22, signal série R20.



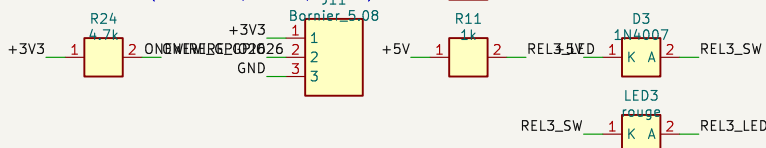
3x HC-SR04 (5V), mode mono-broche TRIG=ECHO (sensor_ultrasonic.cpp).
Echo 5V ramené à 3V3 par pont 1k/2k sur chaque canal.



AIR : DHT11 (option DHT22, -DUSE_DHT22) – pull-up 10k.



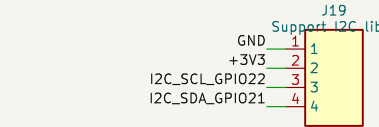
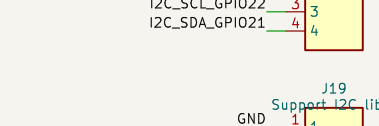
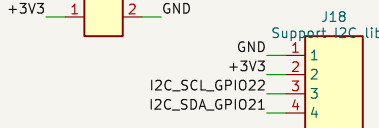
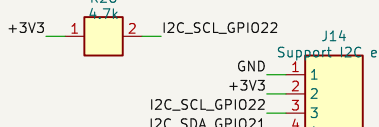
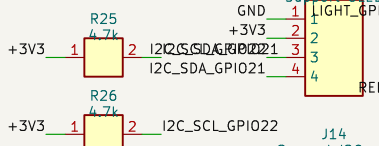
EAU : DS18B20 (1-Wire, pull-up 4.7k).



LUMIERE : LDR déportée + 10k (GPIO34 = entrée seule).



I2C : OLED SSD1306 0x3C + header extension (DS3231 si besoin).



Source de verite : ffp5cs/include/pins.h (BOARD_WROOM) via pinmap.json
Genere par hardware/ffp5cs-wroom-prod/generator/generate.py
salle aeree n3
Sheet: /
File: ffp5cs-wroom-prod.kicad_sch

Title: ffp5cs wroom-prod – carte porteuse bi-module ESP32 (DevKit V1 / S

Size: A3 Date: 2026-07-07 Rev: 0.6
KiCad E.D.A. 8.0.9 Id: 1/1